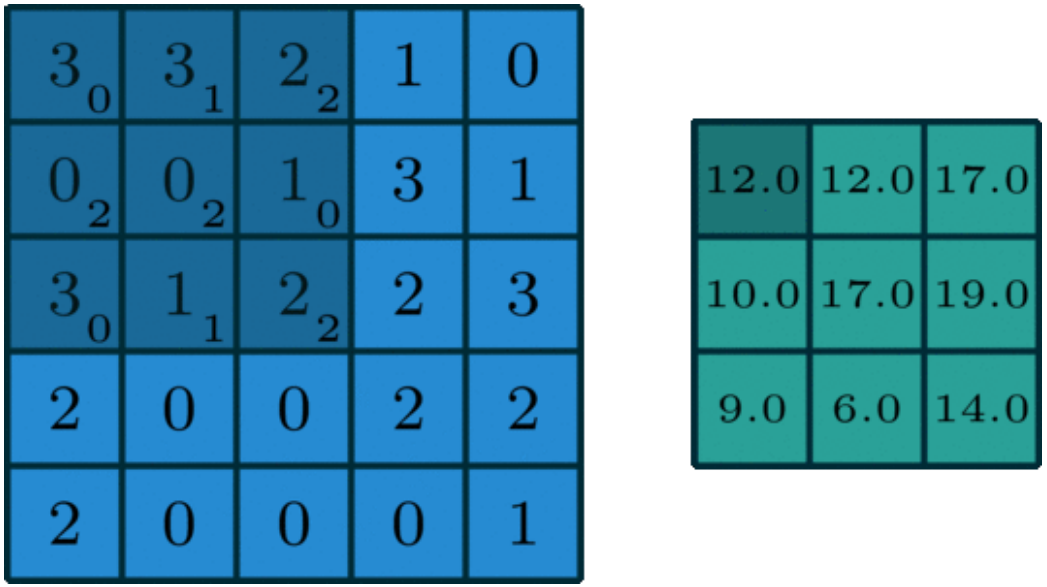
COMS20017 – Algorithms & Data



Convolutions

November 2025

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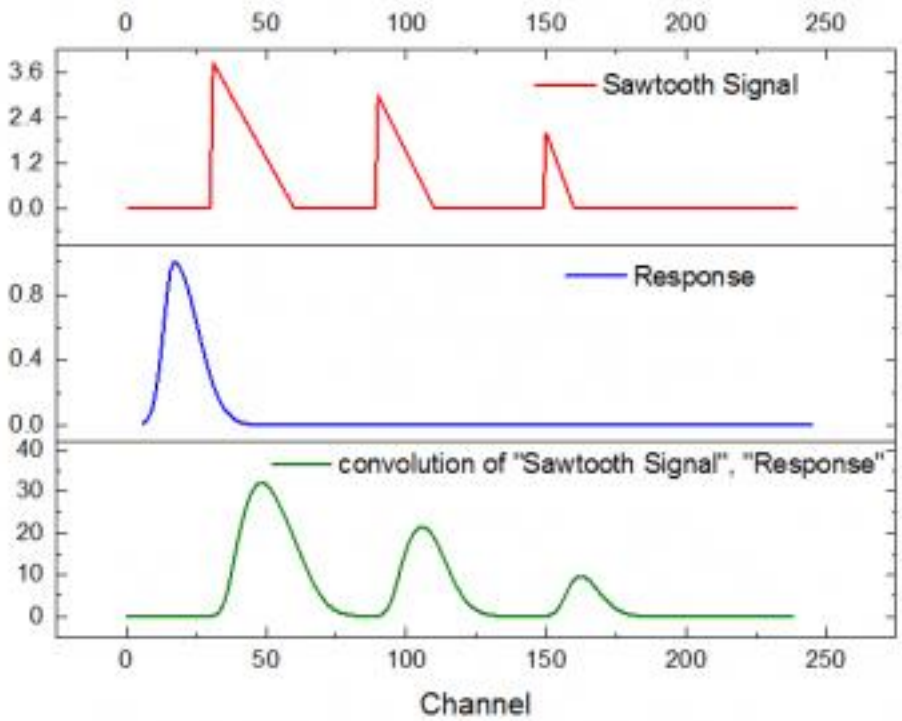
Lecture MM-08

Next in Data



Feature Selection and Extraction

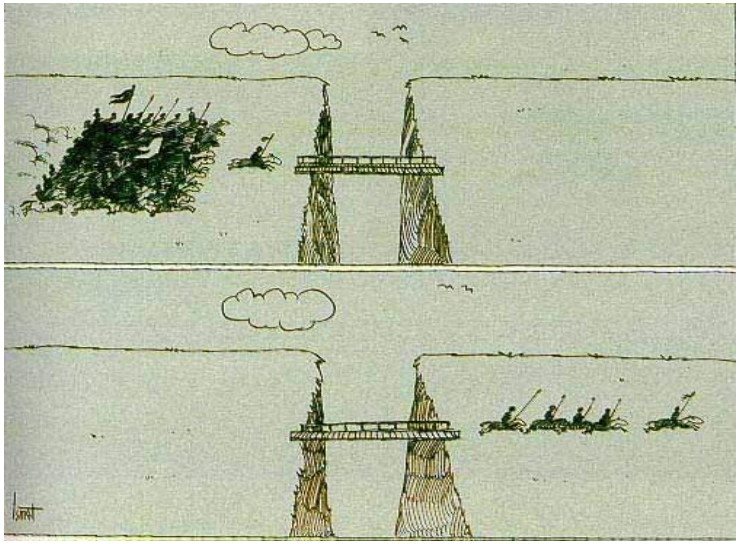
- Signal basics and Fourier Series
- 1D and 2D Fourier Transform
- Another look at features
- PCA for dimensionality reduction
- **Convolutions**

https://www.originlab.com/doc/Origin-Help/Convolution

Spatial Filtering

We can filter signals and symbols in the spatial/time domain:

- introduce some form of enhancement
  - remove noise/outliers
  - smoothing/averaging out detail
  - sharpening/highlighting detail
- prepare for next processing stage
  - feature extraction



Filters are also referred to as *kernels* or *masks*.

Spatial Filtering

Many spatial filters are implemented with ***convolution*** masks.

To do convolution, we need to know about ***neighbourhoods***.

Symbolic Data

ATAGACATGGC .....

neighbours of G?

1D signal data

3	2	4	4	2	6	.....

2D signal data

3	2	4	4	2	6
3	4	5	4	3	6
4	2	5	4	3	3
3	0	4	1	2	6
3	2	4	5	2	6

.....

Convolution mask is applied to each signal sample and its neighbourhood.

Convolution

$$g(x) = h(x) * f(x)$$

Convolution Kernel $h(x)$

1	2	1

Signal $f(x)$

1	1	2	2	1	1	0	1	2	2

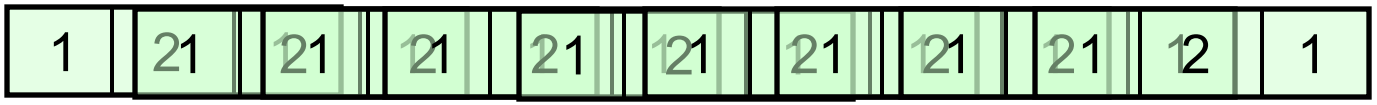
Filter Response $g(x)$

	$\frac{5}{4}$								

Convolution

$$g(x) = h(x) * f(x)$$

Convolution Kernel $h(x)$



Signal $f(x)$

1	1	2	2	1	1	0	1	2	2

Filter Response $g(x)$

☹	$\frac{5}{4}$	$\frac{7}{4}$	$\frac{7}{4}$	$\frac{5}{4}$	$\frac{3}{4}$	$\frac{2}{4}$	$\frac{4}{4}$	$\frac{7}{4}$	☹

Convolution

- f is the signal, h is the convolution filter
  - h has an origin

Example 1D kernel $\frac{1}{5}$

-1	3	-1

  - Normalization factor (sum of the absolute values of the filter) is also part of the filter.

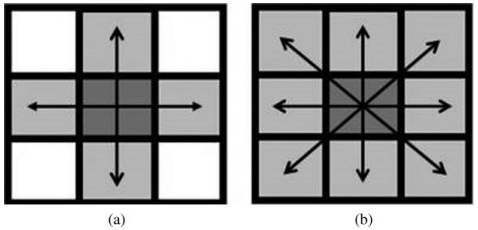
$$g = f(t) * h(t)$$

$$g = \int_{-\infty}^{\infty} f(t-\tau)h(\tau)d\tau = \int_{-\infty}^{\infty} f(\tau)h(t-\tau)d\tau$$

- The discrete version of convolution is defined as:

$$g(x) = \sum_{m=-s}^{s} f(x-m)h(m) \qquad \text{for } s \geq 1$$

2D Spatial Filtering - Connectivity



Determine the connectivity for ***neighbourhoods:***

2D signal data

4–connectivity

3	2	4	4	2	6
3	4	5	4	3	6
4	2	5	4	3	3
3	0	4	1	2	6
3	2	4	5	2	6

8–connectivity.

3	2	4	4	2	6
3	4	5	4	3	6
4	2	5	4	3	3
3	0	4	1	2	6
3	2	4	5	2	6

2D Convolution

- The discrete version of 2D convolution is defined as

$$g(x,y) = \sum_{m=-1}^{1} \sum_{n=-1}^{1} f(x-m, y-n)h(m,n)$$

Shorthand form:

$$g = f * h$$

Convolution symbol

h

	-1	0	1
-1	-1	0	1
0	-2	0	2
1	-1	0	1

f

	$y-1$	y	$y+1$	
$x-1$	43	12	61	
x	44	45	60	
$x+1$	43	50	61	

$$\begin{aligned}
& f(x+1, y+1)h(-1,-1) \\
& + f(x+1, y)h(-1,0) \\
& + f(x+1, y-1)h(-1,1) \\
& + f(x, y+1)h(0,-1) \\
& + f(x, y)h(0,0) \\
& + f(x, y-1)h(0,1) \\
& + f(x-1, y+1)h(1,-1) \\
& + f(x-1, y)h(1,0) \\
& + f(x-1, y-1)h(1,1)
\end{aligned}$$

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2D Correlation

- The discrete version of 2D correlation is defined as

$$g(x,y) = \sum_{m=-1}^{1} \sum_{n=-1}^{1} f(x+m, y+n)h(m,n)$$

h

	-1	0	1
-1	-1	0	1
0	-2	0	2
1	-1	0	1

f

		$y-1$	y	$y+1$	
$x-1$		43	12	61	
x		44	45	60	
$x+1$		43	50	61	

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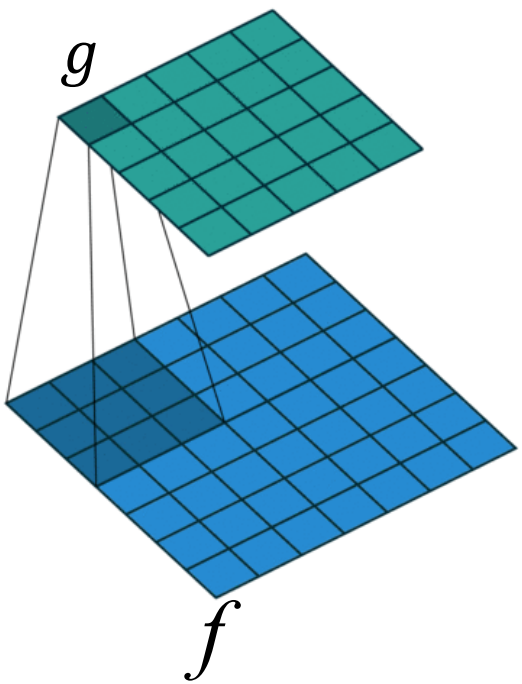
Correlation=Convolution when kernel is symmetric under 180° rotation, e.g.

1	2	1

Spatial Filtering using 2D Convolution (actually Correlation)



Simple example

f

3_0	3_1	2_2	1	0
0_2	0_2	1_0	3	1
3_0	1_1	2_2	2	3
2	0	0	2	2
2	0	0	0	1

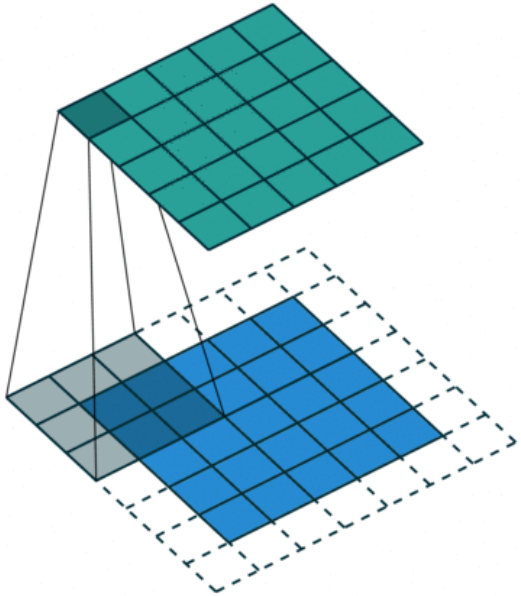
h

0	1	2
2	2	0
0	1	2

g

12.0	12.0	17.0
10.0	17.0	19.0
9.0	6.0	14.0

Use padding for same size result



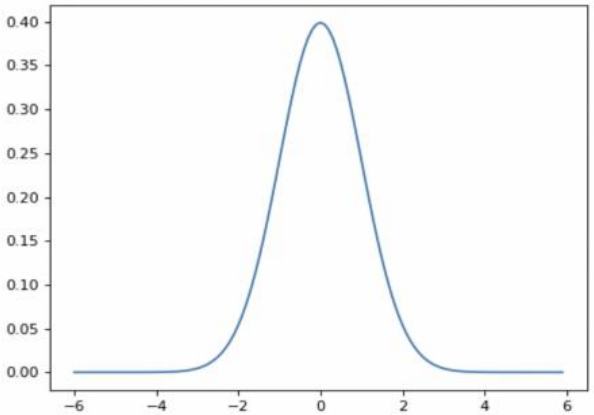
https://towardsdatascience.com/intuitively-understanding-convolutions-for-deep-learning-1f6f42faee1

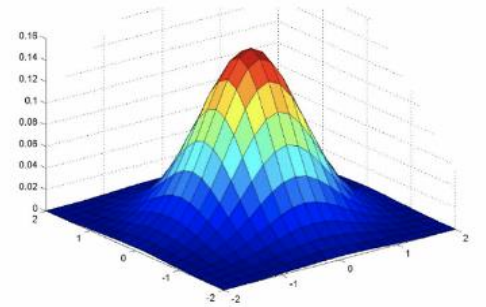
Example: Spatial Low/High Pass Filtering

- 1D: turning the treble/bass knob down on audio equipment!
- 2D: smooth/sharpen image

$$\frac{1}{9}\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \qquad \frac{1}{16}\begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

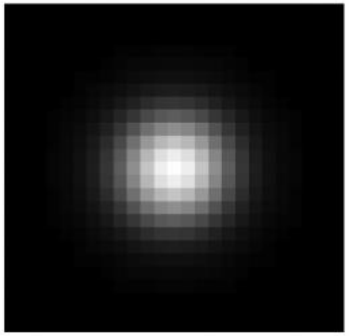
Gaussian Low Pass Filter





Gaussian kernel

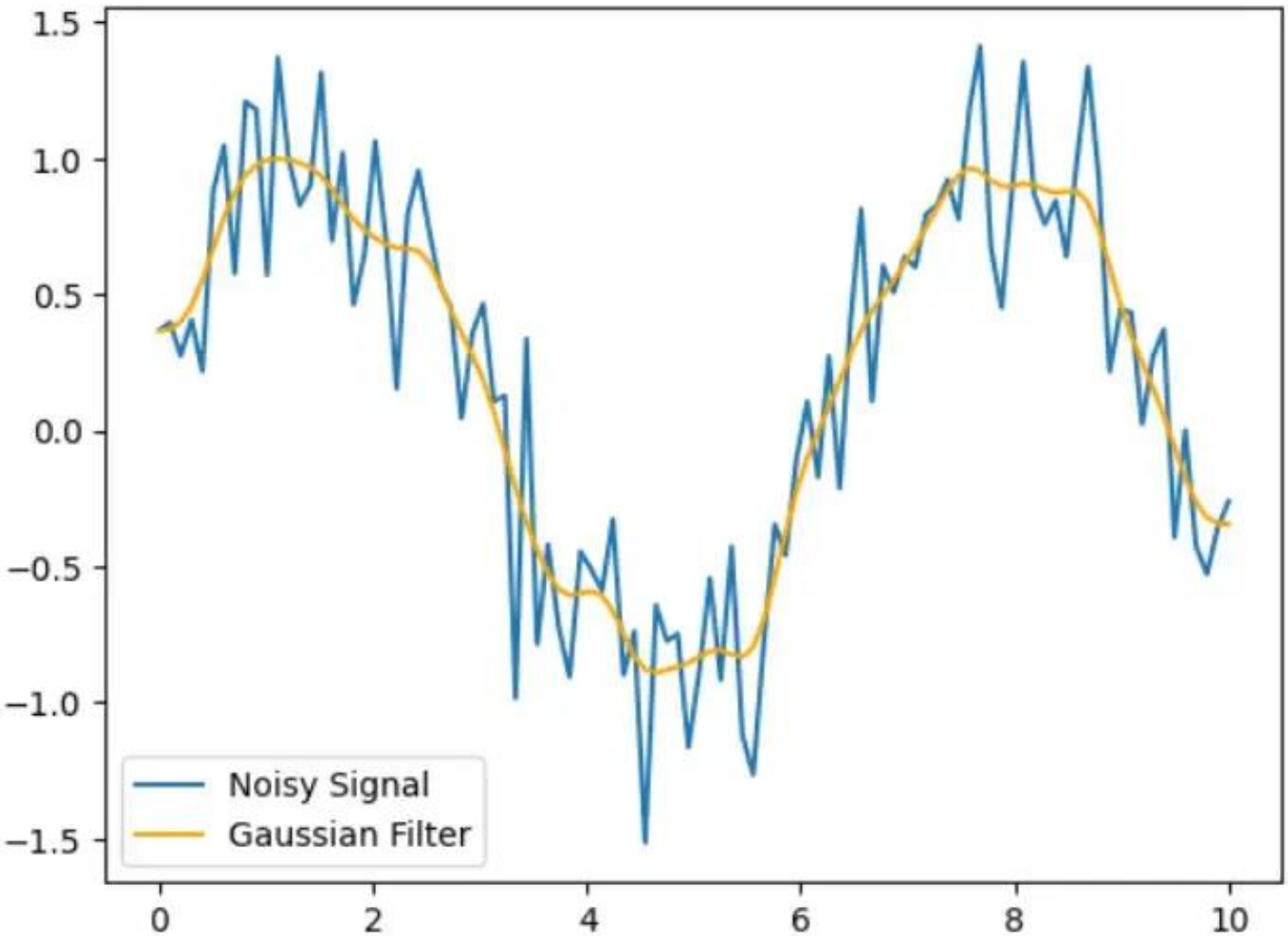
$$G_\sigma = \frac{1}{2\pi\sigma^2} e^{-\frac{(x^2+y^2)}{2\sigma^2}}$$



Example:

$$\frac{1}{16}\begin{pmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{pmatrix} = \frac{1}{16}\begin{pmatrix} 1 \\ 2 \\ 1 \end{pmatrix}\begin{pmatrix} 1 & 2 & 1 \end{pmatrix}$$

Example: Gaussian Low Pass Filter in 1D



Example: Edge Features

$$f(x, y)$$

- Edges occur in images where there is discontinuity (or change) in the intensity function.

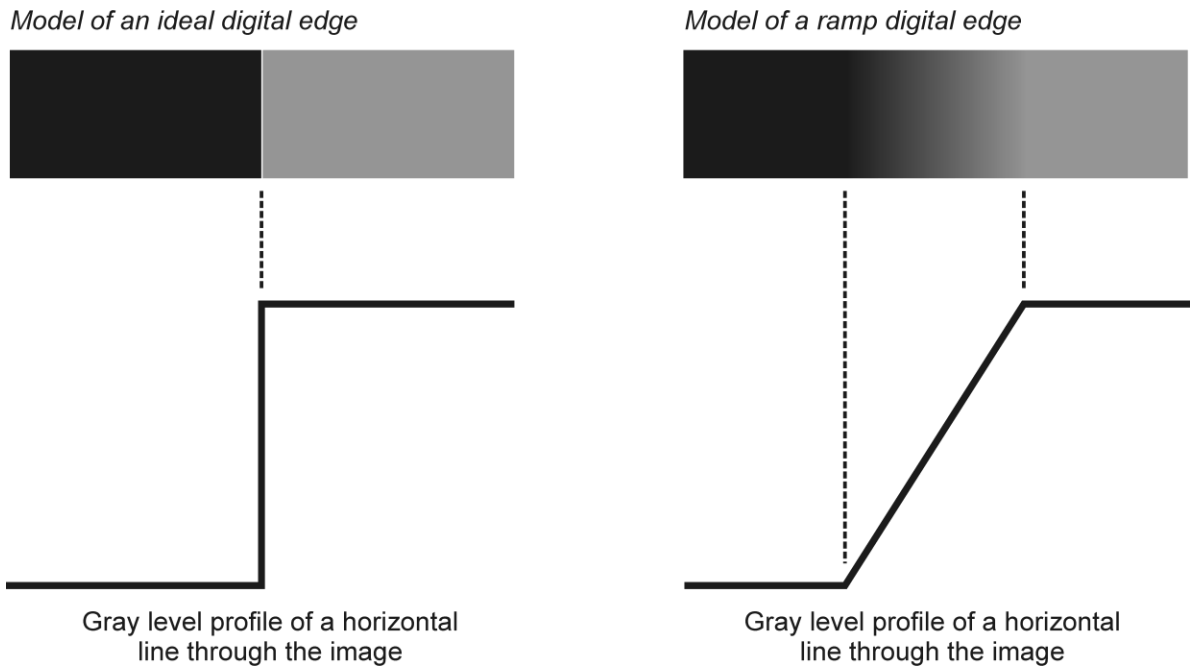
- Gradient points in the direction of most rapid change in intensity

- Biggest change → derivative has maximum magnitude.

$$\|\nabla f\| = \sqrt{\left[\frac{\partial f}{\partial x}\right]^2 + \left[\frac{\partial f}{\partial y}\right]^2}$$

Edge detection: basic concept

- Edge: a boundary between two image regions having distinct characteristics according to some feature (e.g., gray level, color, or texture).
- In grayscale 2D images: a significant variation of the intensity function across a portion of the image.



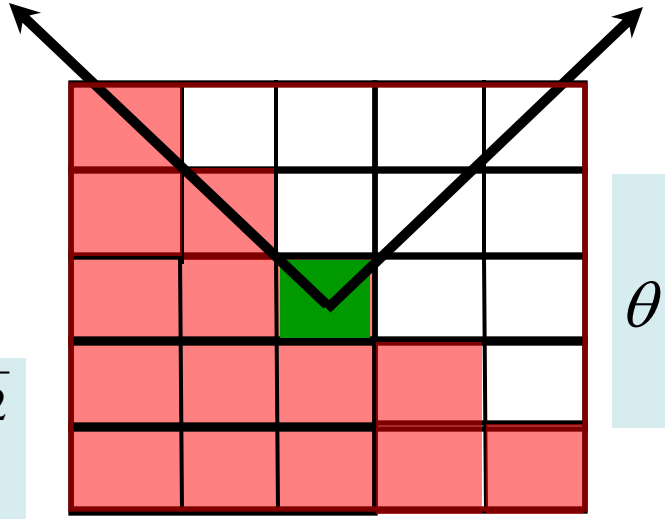
Edge Measures

Edge direction

$$\phi = \theta - \frac{\pi}{2}$$

Edge magnitude

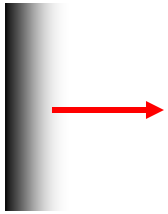
$$\|\nabla f\| = \sqrt{\left[\frac{\partial f}{\partial x}\right]^2 + \left[\frac{\partial f}{\partial y}\right]^2}$$



gradient direction

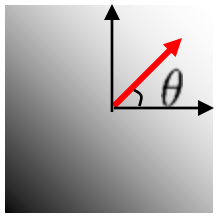
$$\theta = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$$

$$\nabla f = \left[\frac{\partial f}{\partial x}, 0\right]$$



$$\nabla f = \left[0, \frac{\partial f}{\partial y}\right]$$

$$\nabla f = \left[\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right]$$



Sobel Edge Detector

- 2D gradient measurement in two different directions.
- Uses these $3x3$ convolution masks:

-1	0	+1
-2	0	+2
-1	0	+1

$*$

$\frac{\partial f}{\partial x} =$

+1	+2	+1
0	0	0
-1	-2	-1

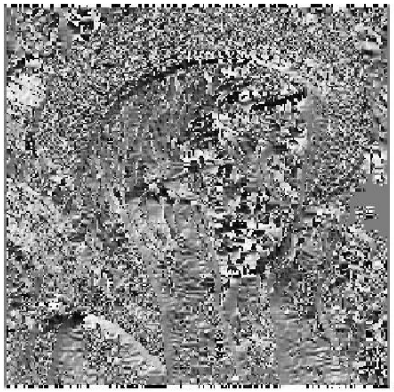
$*$

$\frac{\partial f}{\partial y} =$

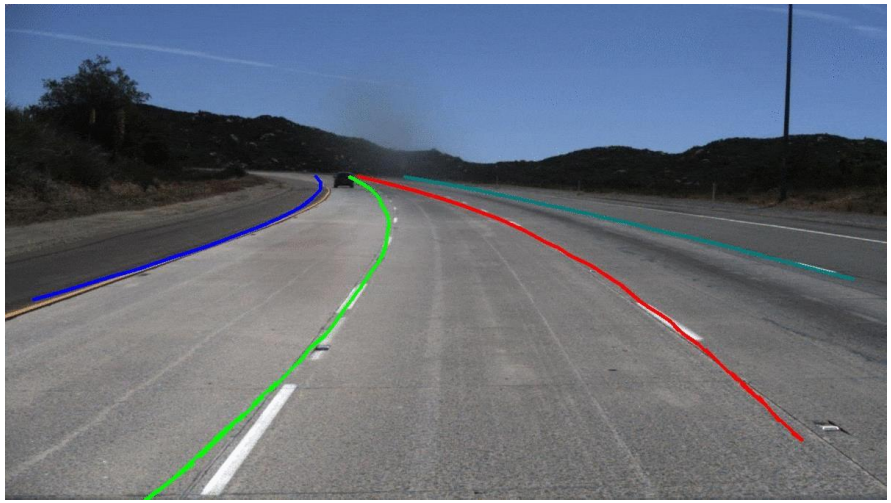
Thresholded result

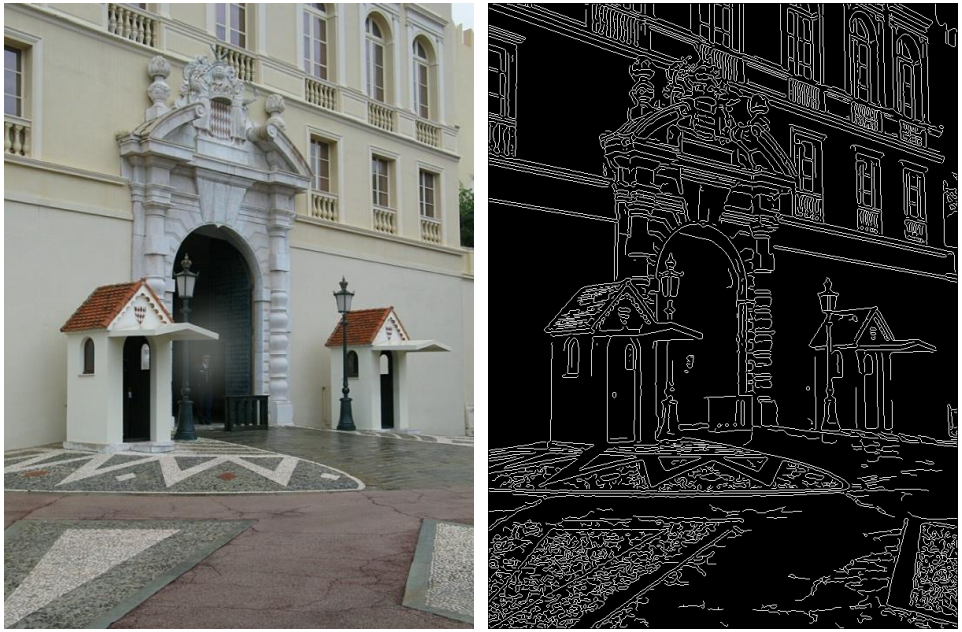
$\|\nabla f\|$

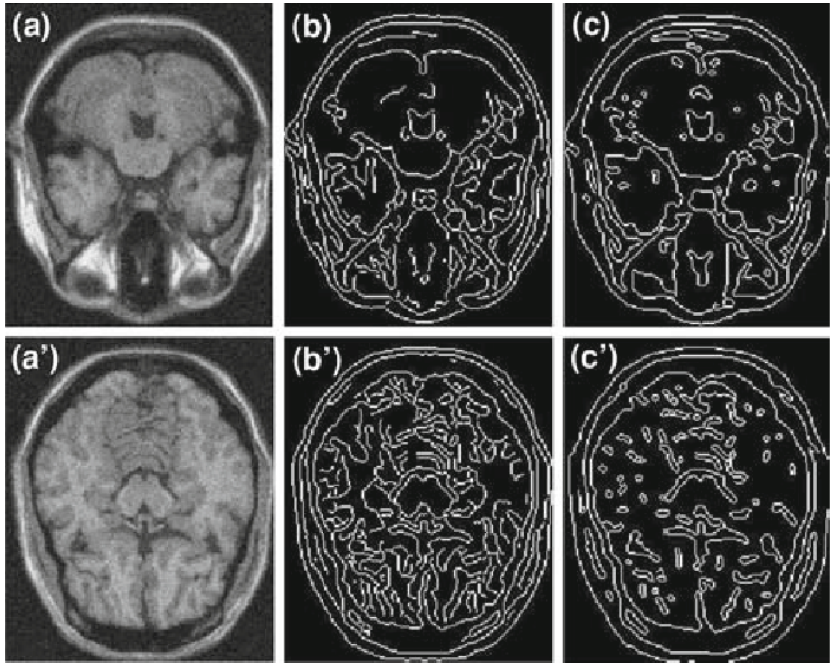
θ



Examples

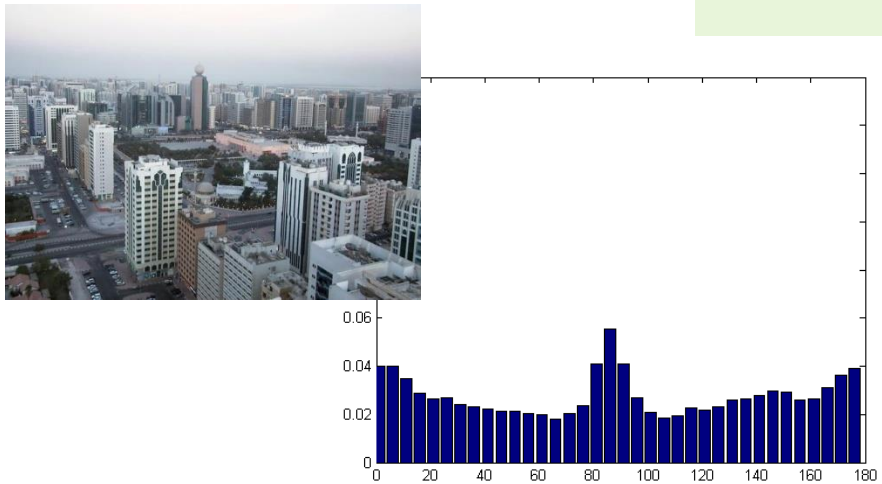


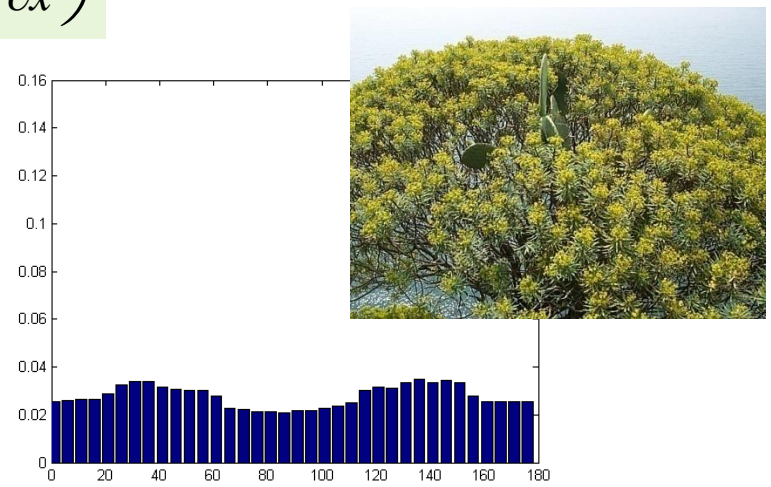


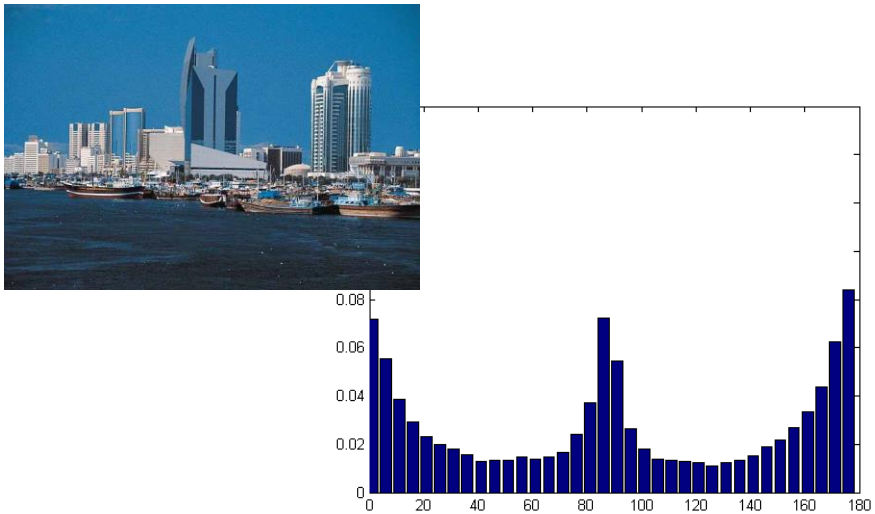


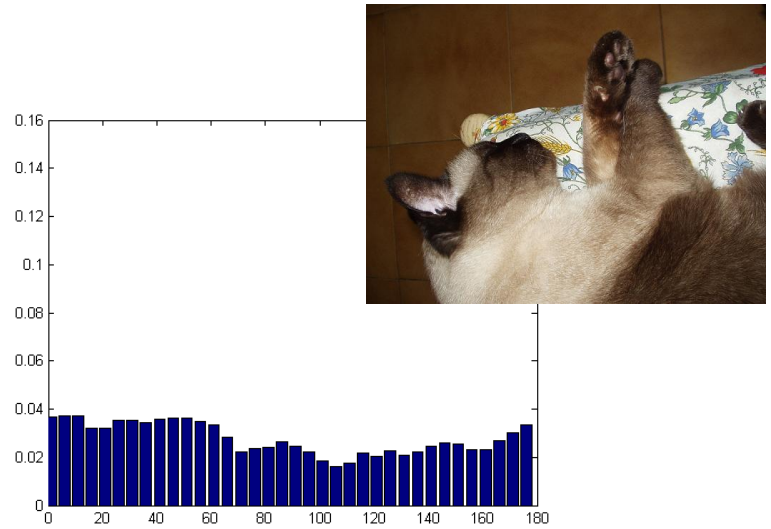
Histogram of Edge Gradient Directions

$$\theta = \tan^{-1}\left(\frac{\partial f}{\partial y} / \frac{\partial f}{\partial x}\right)$$









Matlab: Sobel Edge Detection

```
% Sobel edge detection
A = imread('romina.gif');
fx = [-1 0 1; -2 0 2; -1 0 1]
fy = [1 2 1; 0 0 0; -1 -2 -1]
gx = conv2(double(A),double(fx))/8;
gy = conv2(double(A),double(fy))/8;
mag = sqrt((gx).^2+(gy).^2);
ang = atan(gy./gx);
figure; imagesc(mag); axis off; colormap gray
figure; imagesc(ang); axis off; colormap gray
```

See unit github page for code in Python.

Spatial/Frequency Domain Filtering

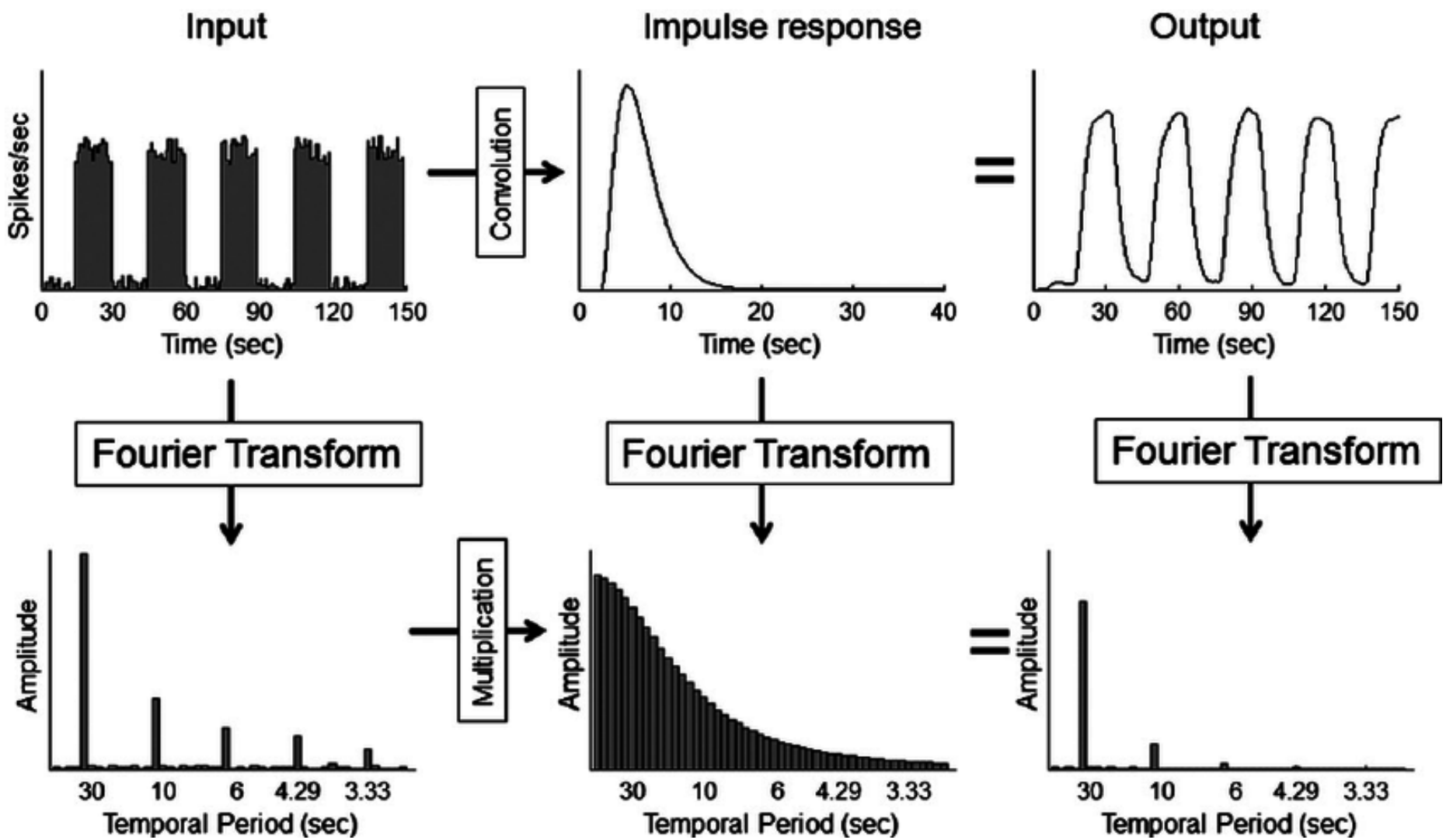
- Convolution Theorem:

Convolution in the spatial domain
is equivalent to
multiplication in the frequency domain
and vice versa

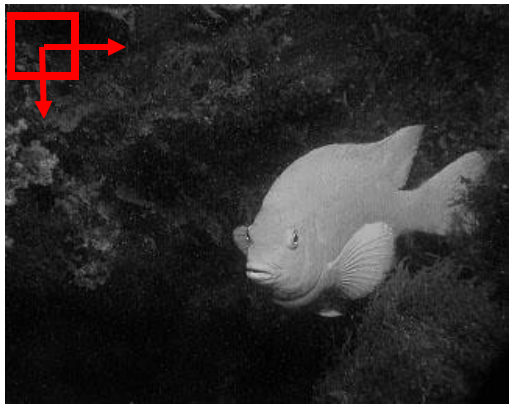
$$g(x) = f(x) * h(x) \quad \Longleftrightarrow \quad G(u) = F(u)\,H(u)$$

$$g(x) = f(x)\,h(x) \quad \Longleftrightarrow \quad G(u) = F(u) * H(u)$$

Example: Convolution in SD is Multiplication in FD

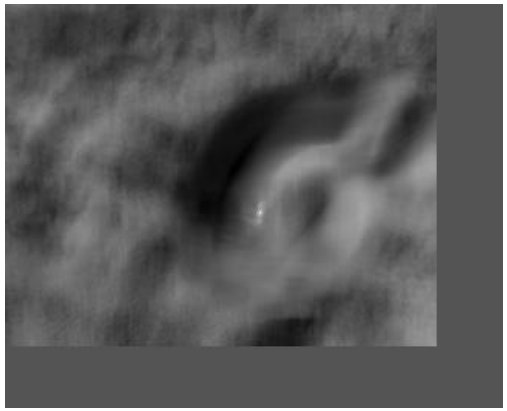


Correlation Example: Fish parts!



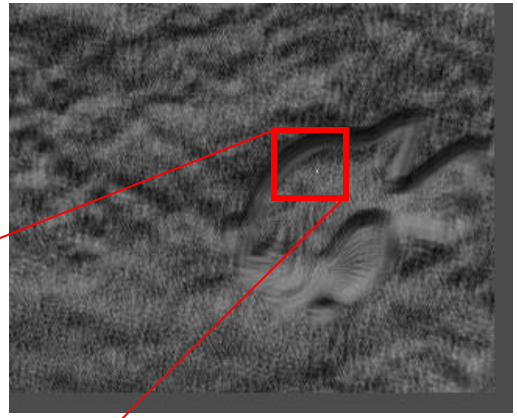
u

v_{eye}

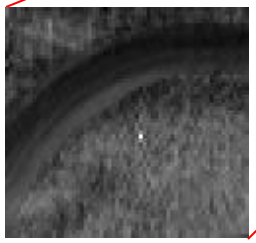


ρ_{eye}

v_{back}



ρ_{back}



Template Matching

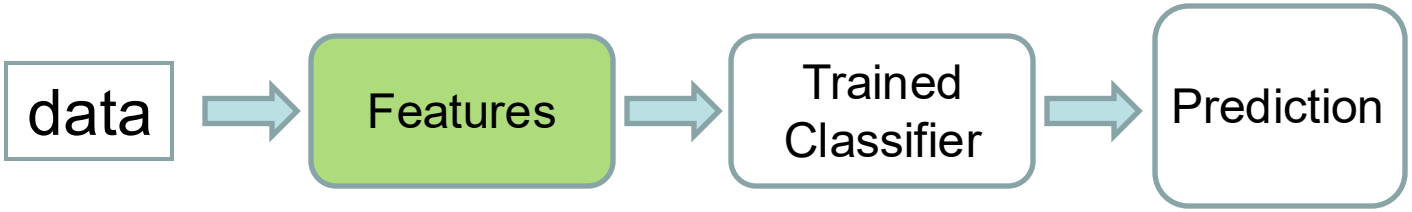
Convolution Exercise

$$P = \begin{pmatrix} -1 & 4 & -1 \\ 4 & 10 & 4 \\ -1 & 4 & -1 \end{pmatrix} \qquad Q = \begin{pmatrix} 5 & 3 & 0 & 0 & 7 \\ 3 & 3 & 2 & 1 & 4 \\ 6 & 5 & 1 & 1 & 6 \\ 6 & 3 & 4 & 1 & 4 \\ 4 & 4 & 5 & 0 & 4 \end{pmatrix}$$

Apply convolution filter P to matrix Q at location (2,3).

Answer = ?

Well Done for Lasting This Far!



Feature Selection and Extraction

- Signal basics and Fourier Series
- 1D and 2D Fourier Transform
- Another look at features
- PCA for dimensionality reduction
- Convolutions